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SURVEY

Explainable Artificial Intelligence (XAI): Concepts, Applications, Challenges, and Future Perspectives

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ABSTRACT The aim of explainable artificial intelligence (XAI) is to address the black-box problem in high-stakes applications. However, transparency alone does not guarantee trust. This review examines a critical paradox in XAI research. While explanation methods can generate insights, three main challenges limit their effectiveness. Firstly, adversarial manipulations can exploit explanations by creating new attack surfaces with over ninety percent success while preserving model accuracy. Secondly, evaluation practices remain primarily computational. Only twenty-six percent of user studies follow human-centered protocols and fewer than twenty-three percent involve domain experts. Thirdly, regulatory requirements, such as the GDPR right to explanation, lack clear technical implementations, complicating compliance. We analyzed the literature across finance, healthcare, and cybersecurity and found that current research emphasizes algorithmic innovation over practical deployment. Moving toward reliable AI requires shifting from simple explanation methods (XAI 1.0) to systems that are aligned with human understanding, resistant to adversarial attacks, and compliant with legal requirements (XAI 2.0). This review provides guidance on key technical advances, evaluation strategies and regulatory clarifications necessary for deployment. trustworthy AI.

INDEX TERMS Ante-hoc, explainable artificial intelligence (XAI), human-centered XAI, interpretable machine learning, trustworthy AI, post-hoc.

I. INTRODUCTION

Artificial intelligence (AI), especially deep learning, has changed the technology of the twenty-first century in a big way. These systems have changed many industries and scientific fields, and they are better than human experts at recognizing images and speech [1], running autonomous systems, and diagnosing complicated medical conditions [2]. Deep neural networks are able to do this by finding complex, hierarchical patterns in huge datasets with millions or billions of parameters. But this ability to predict comes at a cost: as

models get more complicated, it's harder to understand how they make decisions. In artificial intelligence, the black box problem describes systems that generate correct outputs without revealing how those decisions are made [4], [5]. As AI is increasingly being used in areas such as healthcare, finance, criminal justice and transportation where safety is crucial, this lack of transparency has become a significant concern, since errors in algorithms may have serious consequences [6]. A lack of transparency leads to a loss of trust and accountability, and makes it difficult to verify outcomes [7]. Furthermore, regulations such as the GDPR entitle individuals affected by automated decisions to an explanation [8]. The demand for interpretability, driven by these legal and ethical demands,

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has led to the rapid growth of explainable AI (XAI). The aim of XAI is to make AI decisions more understandable, reduce biases, and support the development of reliable and fair systems [9], [10], [11], [12].

Current XAI surveys typically treat explainability as primarily a technical challenge, with the focus being on generating interpretable outputs. However, failures observed in real-world applications present a more complex picture. For example, if a medical AI system provides accurate diagnoses but clinicians distrust its explanations, the issue may lie not only in the underlying algorithms, but also in the system's insufficiently human-centered design. These considerations show that building trustworthy AI requires more than just sophisticated explanation techniques. This review addresses gaps in the existing literature by making three key contributions. Firstly, we demonstrate that adversarial attacks significantly reduce the reliability of explanation methods [13], [14], yet defense strategies are not widely adopted. Secondly, we present evidence suggesting that current evaluation practices prioritize computational metrics over human interpretability. This helps to explain why systems that perform well in controlled settings often fail in deployment. Lastly, we examine the challenges that organizations face in meeting regulatory expectations when they cannot determine whether their explanations satisfy legal standards, demonstrating that regulatory and technical requirements are not aligned, which makes practical implementation difficult.

A. FUNDAMENTAL CONCEPTS IN EXPLAINABLE ARTIFICIAL INTELLIGENCE

XAI refers to the techniques and methods that make the behavior and predictions of AI systems understandable to humans. Unlike traditional black-box models, which provide predictions without justification, XAI systems aim to provide transparent reasoning, allowing users to understand, trust and appropriately rely on AI decisions. The term explainable AI formally emerged in the mid-2010s, although research on interpretable models predates this terminology. Significant momentum was gained in the field following DARPA's XAI program (2016–2021), which established foundational frameworks for distinguishing between models that are interpretable by design (ante-hoc methods) and techniques for providing explanations after the fact (post-hoc) that can be applied to black-box systems [128]. Early foundational work in XAI includes decision tree visualization, rule extraction from neural networks and sensitivity analysis [10], [41]. Pivotal developments in model-agnostic post-hoc explanations were marked by Schlegel et al.'s LIME (2016) and Lundberg and Lee's SHAP (2017), enabling practitioners to generate local or global explanations for any classifier [35], [36], [37]. At the same time, gradient-based attribution methods such as Grad-CAM [54] emerged to improve the interpretability of deep learning in computer vision. These seminal contributions have laid the methodological groundwork for

contemporary XAI research. The need for XAI arises from multiple converging factors. Firstly, regulatory requirements such as EU's General Data Protection Regulation (GDPR) [8] and the proposed AI Act mandate transparency in automated decision-making processes. Secondly, in high-stakes domains such as healthcare diagnosis, credit assessment and criminal justice, practitioners must understand AI reasoning in order to identify potential biases, validate clinical or legal compliance and maintain accountability [6]. Thirdly, psychological research shows that users must understand when to rely on and when to override AI recommendations in order to trust it appropriately [7]. Without explainability, even highly accurate models risk being misused, underutilized, or deployed incorrectly when users cannot assess their reliability in new situations. XAI is distinct from, yet related to, concepts such as interpretability, transparency and trustworthiness. While interpretability refers to the extent to which humans can understand the cause of a decision, transparency describes the degree to which a model's internal mechanisms are visible. Trustworthiness encompasses broader considerations, including robustness, fairness and privacy, with explainability serving as a critical component. This review takes an inclusive approach, examining XAI methods across the spectrum, from intrinsically interpretable models to sophisticated post-hoc explanation techniques. It recognizes that different stakeholders and application contexts require different explanation paradigms.

The important contributions of this paper to the literature are summarized below.

- We provide a comprehensive and structured taxonomy of XAI by systematically organizing explanation methods across multiple dimensions, including model dependency, scope of explanation, and timing (ante-hoc versus post-hoc), offering a unified conceptual framework for both researchers and practitioners.
- We present a domain-oriented analysis of XAI applications in high-stakes sectors such as finance, healthcare, and cybersecurity, highlighting domain-specific requirements, user roles, and deployment constraints that are often overlooked in existing surveys.
- We conduct an in-depth examination of security and robustness challenges in XAI, including adversarial attacks on explanations, fairwashing, and explanation manipulation, and critically discuss emerging defense and reliability assessment strategies.
- We systematically analyze current evaluation practices for XAI, demonstrating that the majority of studies rely on computational metrics while underutilizing human-centered evaluation frameworks, and we identify key limitations related to scalability, standardization, and user trust.
- We bridge technical, human-centered, and regulatory perspectives by explicitly discussing the gap between existing XAI methods and legal requirements such as the GDPR right to explanation, thereby clarifying practical challenges in real-world deployment.

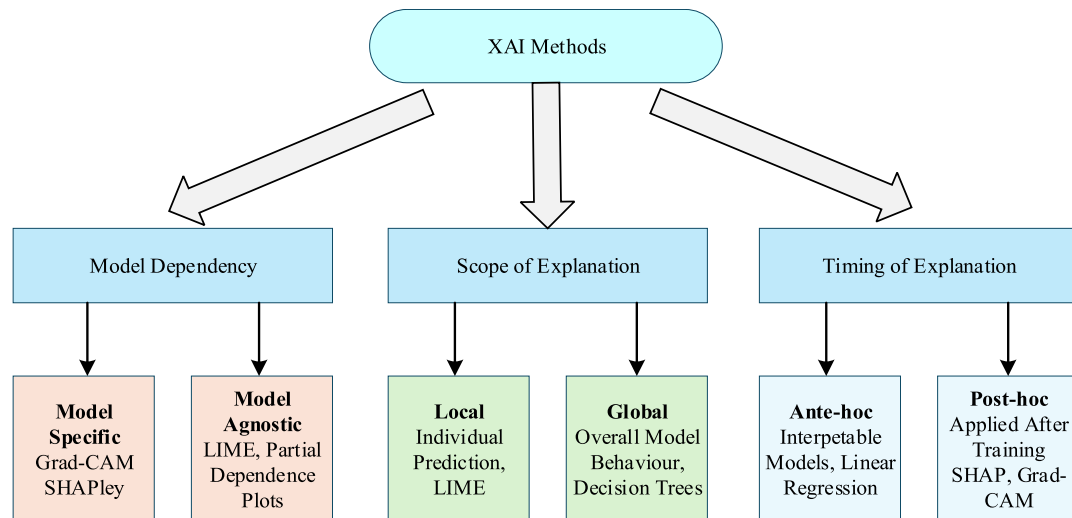


FIGURE 1. Conceptual classification of XAI methods based on key taxonomic axes such as model dependency, coverage, and timing of explanation.

- We synthesize recent advances and open challenges to outline a forward-looking research roadmap toward XAI 2.0, emphasizing the need for human-aligned, adversarially robust, and regulation-aware explainability systems, particularly in the context of foundation models and multimodal AI.

The paper is organized as follows. Section II outlines the scope of the review and the methodology employed. Section III provides an overview of core XAI principles and major techniques. Applications in the fields of finance, healthcare and cybersecurity are presented in Section IV. Section V discusses explanation robustness, including adversarial risks and defense approaches. Section VI evaluates assessment practices, emphasizing the need for human-centered frameworks. Section VII highlights ongoing technical and ethical challenges, as well as potential research directions. Section VIII acknowledges the limitations of the review. Section IX concludes by presenting the main insights and providing a roadmap for future XAI research.

II. REVIEW SCOPE AND APPROACH

Three main research questions (RQ1–RQ3) guide this review. RQ1 focuses on identifying the main XAI methods and examining how they are systematically categorized across different model types and explanation paradigms. RQ2 investigates how XAI applications address domain-specific challenges, particularly in high-stakes contexts such as finance, healthcare, and cybersecurity. RQ3 examines the potential limitations and risks associated with deploying XAI systems, including issues related to security, robustness, and regulatory compliance.

The formulation of these research questions was informed by an iterative analysis of existing XAI surveys and recent application-driven studies. While prior works have extensively addressed individual aspects such as the definition

and classification of XAI methods, our review identified that these dimensions are often examined in isolation, without being systematically linked to deployment challenges, evaluation practices, and regulatory considerations. In this context, RQ1 does not aim to introduce an entirely novel taxonomy; instead, it synthesizes widely accepted classification dimensions into a unified and deployment-oriented framework that serves as a foundation for the subsequent analysis in RQ2 and RQ3. This integrative formulation enables a coherent examination of how methodological choices in XAI influence real-world applicability, security, and compliance in high-stakes settings. To position our contributions within the broader literature, Table 1 compares the scope of this review with that of earlier surveys. This demonstrates that, although previous studies have investigated individual aspects of XAI, such as technical methods, applications in specific sectors and ethical concerns, only a limited number have considered these dimensions together in an integrated manner.

To address these research questions, we have adopted a structured thematic synthesis approach to comprehensively cover the rapidly evolving field of XAI. Although our approach is informed by systematic review principles, we do not strictly adhere to formal protocols such as PRISMA. This is because XAI research is dynamic, and it is important to incorporate recent developments, including preprints and technical reports that have not yet completed peer review. This methodological choice allows us to identify emerging trends in foundation model interpretability and address current deployment issues that would be excluded by rigid temporal or publication-type constraints. Our literature search targeted four major academic databases, which we selected for their comprehensive coverage of the field. IEEE Xplore and the ACM Digital Library for computer science and engineering publications; Scopus for

broad, multidisciplinary coverage; and Arxiv for emerging research and preprints. The search covered publications from 2018 to 2025, focusing on the period during which XAI transitioned from a specialized research area to a critical component of responsible AI deployment. Search queries combined core terminology ('explainable AI', 'XAI', 'interpretable machine learning', 'interpretability', 'transparency') with domain-specific keywords related to methods (e.g. 'LIME', 'SHAP', 'attention mechanisms'), applications (e.g. 'healthcare', 'finance', 'cybersecurity'), and cross-cutting concerns (e.g. 'fairness', 'robustness', 'evaluation', 'human-centredness'). Boolean operators were used to systematically combine these terms, and an English-language constraint was applied to ensure consistency of analysis. The initial search yielded a substantial body of literature that was potentially relevant. The screening process proceeded in multiple stages. First, the titles and abstracts of the literature were reviewed to identify studies that addressed XAI methods, applications, evaluation frameworks, security and robustness considerations, or deployment challenges directly. Studies focusing exclusively on general machine learning without considering explainability, or providing only speculative commentary without making any empirical or methodological contributions, were excluded at this stage. Duplicate articles appearing across multiple databases were identified and removed. The remaining articles were then subjected to a full-text review, which applied stricter eligibility criteria. In order to be considered, studies had to provide substantive empirical evidence, propose or evaluate specific XAI techniques, or offer critical analysis relevant to our research questions. Articles lacking sufficient methodological detail or not directly contributing to the understanding of XAI systems were excluded. We systematically extracted information from the included articles and organized it around our three research questions. The following key data points were included: (1) characteristics of the XAI method (ante-hoc vs. post-hoc; model-specific vs. model-agnostic; local vs. global scope), (2) the application domain and specific use cases, (3) the evaluation approaches employed (computational metrics; human-centred assessment; or hybrid methods), and (4) considerations relating to security, robustness, and regulatory compliance. Articles were classified into thematic categories through iterative discussion among the authors. Any disagreements that arose during classification were resolved through consensus and reference to established XAI taxonomies in prior surveys. This approach was both structured and flexible, enabling us to capture the technical diversity of XAI methods and the contextual nuances of their deployment across high-stakes domains. Figure 2 provides a schematic overview of our literature search and selection process.

III. CONCEPTS AND TAXONOMIES OF XAI

The understanding of XAI requires engagement with several taxonomic dimensions that do not always align in straightforward ways. When selecting an explanation method,

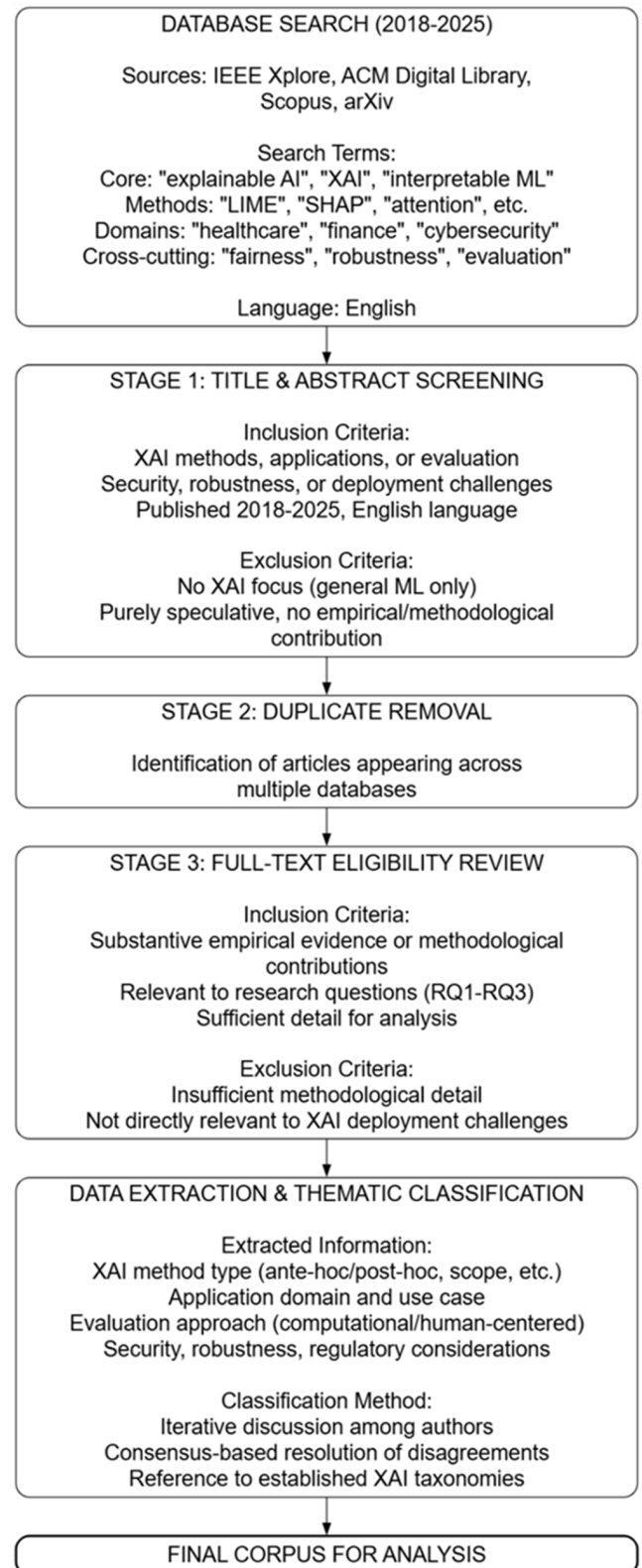


FIGURE 2. Schematic overview of literature search and selection process.

practitioners must consider the interdependency of their choices. Model-specific approaches offer high fidelity, but cannot be applied across different models. In contrast,

TABLE 1. Difference between our survey and other survey papers of a similar nature.

Ref	Conceptual Framework	Explainability Methods (ante-hoc & post-hoc)	Application Areas	Security, Ethical, and Legal Aspects	Human-Centered XAI and User Trust
[18]	√	√	–	–	–
[19]	√	√	–	–	–
[3]	√	–	–	√	–
[20]	–	√	√	–	–
[21]	–	–	√	√	–
[22]	–	–	√	√	–
[23]	–	–	√	–	–
[24]	√	√	–	–	–
[5]	√	–	–	√	√
Our study	√	√	√	√	√

model-agnostic methods provide flexibility at the cost of reduced accuracy. Local explanations address the needs of individual users, but they do not consider broader structural biases. Global explanations capture system level patterns. However, they can overload decision makers. On the one hand, ante hoc interpretability ensures fidelity but restricts model capacity. On the other hand, post hoc explanations allow the use of more complex models but may produce approximations that are not fully reliable. These trade-offs mean that no single XAI method is universally optimal. The suitable option is dependent on the application context, the proficiency of the users and whether the primary goal is troubleshooting, regulatory alignment or establishing confidence. This section provides a foundation for understanding these design considerations and their implications.

In order to introduce a certain degree of order to this complex landscape, we have introduced a taxonomy in Table 2 that organizes XAI methods according to a variety of criteria. This framework enables practitioners to identify suitable explanation techniques based on deployment requirements, user expertise, and regulatory constraints. Table 2 forms the basis of the taxonomy employed throughout this analysis. These categories are not merely conceptual distinctions, but reflect the practical choices that professionals must make when implementing XAI systems. The following subsections examine each dimension in detail and discuss how these design decisions affect the quality of explanations, computational requirements, and user comprehension.

XAI is a broad research area focused on making the reasoning processes of AI systems more understandable. This field has grown rapidly and now encompasses a variety of methods developed for different purposes. Clear conceptual models and structured taxonomies are essential to navigate this variety [26]. This section summarizes core XAI principles and highlights influential taxonomies from recent work, offering guidance on selecting the most suitable explanation techniques. Figure 1 provides a visual overview of XAI methods. These are organized around three key factors. The first

factor is the underlying model. The second factor is the scope of the explanation. The third factor is the timing of the interpretability process. This framework illustrates how different approaches occupy distinct positions in the design space, each with its own strengths and limitations. The XAI landscape is diverse, there is no single solution that outperforms all others. Therefore, selecting the right method requires an understanding of how these dimensions interact and shape the resulting explanations. The following subsections provide more detailed exploration of each factor and emphasize the principles that support effective and meaningful XAI practice.

A. FUNDAMENTAL PRINCIPLES OF EXPLAINABILITY

Meaningful explanations strike a balance among various attributes. Interpretability is the ability to understand how a model works on the inside, while explainability is the ability to explain how a model works in terms that people can understand [28]. A radiologist analyzing a diagnostic AI necessitates distinct explanations compared to a data scientist troubleshooting the identical model. Fidelity assesses the extent to which explanations accurately represent authentic decision-making processes [29]. Low-fidelity explanations trick users into thinking they know what's going on, which is worse than not knowing what's going on. Comprehensibility makes sure that people can understand explanations without having to think too hard. An explanation that is completely true to model mechanics but makes no sense to the person who is supposed to use it is not useful. Finally, scope sets apart local explanations that clarify individual choices from global explanations that sum up how the whole model works [30].

B. MODEL-SPECIFIC AND MODEL-AGNOSTIC APPROACHES

A prevalent taxonomy categorizes methods according to their reliance on particular machine learning models [31], [32]. Model-specific methods are designed to take advantage of

TABLE 2. XAI concepts and taxonomies.

Classification Axis	Category	Definition	Advantages	Disadvantages	Sample Methods
Model Dependency	Model-Specific	Designed to exploit only the internal structure of a specific model family.	Offers high fidelity and computational efficiency.	Limited to certain model types.	Decision Trees, Linear Models, Grad-CAM
	Model-Agnostic	It treats the model to be explained as a black box and makes no assumptions about its internal structure.	Can be applied to any machine learning model and is versatile.	May be less accurate than model-specific methods.	LIME, SHAP
Scope of Explanation	Local	It focuses on explaining why the model made a particular decision for a single data point.	Invaluable for end users and decision makers.	Does not summarize the overall behavior of the model.	LIME, Individual SHAP Values
	Global	It aims to summarize the model's behavior and logic across the entire dataset.	Critical for understanding the overall consistency and potential biases of the model.	May be inadequate in explaining single, specific decisions.	Overall SHAP Values, Partial Dependency Plots (PDP)
Timing of Explanation	Ante-hoc	It includes design-interpretable models, which are inherently transparent and interpretable.	Because the explanation and the model are the same, the generated explanations are completely faithful to the model.	Generally, it is not as effective as black-box models in modeling complex relationships.	Linear Regression, Logistic Regression, Shallow Decision Trees
	Post-hoc	After the model's training process is complete, it generates explanations by applying them to an existing black-box model.	Indispensable for increasing the transparency of high-performance but complex models.	Fidelity can be an issue in how faithful the explanation is to the model's internal logic.	LIME, SHAP, Grad-CAM
Post-hoc subcategory	Feature Attribution	It quantifies the importance or contribution of each feature.	Clearly shows which inputs most influence the decision.	May overlook interactions between features.	SHAP Values
	Counterfactual	It specifies the minimum change that must be made to the inputs for the model to make a different decision.	Provides the user with actionable steps on how to change an outcome.	May not always produce intuitive or applicable scenarios.	-
	Example-Based	It uses the most relevant or prototypical examples from the training data to explain the model's decision.	Makes it easier for users to understand by comparing similar situations.	Finding optimal examples can be computationally expensive.	-

the internal structure of certain model families [33]. They usually have better accuracy and are faster at processing. For instance, following decision tree paths shows reasoning clearly and directly [34], and linear model coefficients show the effect of each feature directly. In deep learning, Grad-CAM (Gradient-weighted Class Activation Mapping) makes visual explanations by using heatmaps to show important parts of an image.

Model-agnostic methods look at the input-output relationships of models without looking at their internal structure [31], [32]. This adaptability permits utilization across any machine learning model, which is advantageous in industrial contexts where proprietary models are impervious to scrutiny. Two main methods stand out and should be looked at closely. LIME (Local Interpretable Model-Agnostic) trains simple

surrogate models that make predictions about black boxes in a specific area [35]. SHAP (Shapley Additive exPlanations) uses Shapley values from cooperative game theory to figure out how much each feature contributes [36], [37]. SHAP provides unique feature explanations, but calculating exact values can be computationally demanding. For a model with M features, all 2^M feature combinations must be evaluated. This issue is usually resolved by using approximate methods. Kernel SHAP reduces the number of evaluations required by applying weighted linear regression, typically needing only 2,000 to 5,000 evaluations per explanation. Tree SHAP uses the structure of decision tree ensembles to perform calculations in polynomial time, making the process about 2.5 times faster than Kernel SHAP [38]. These approximation methods involve significant trade-offs.

TABLE 3. Practical guide: Matching XAI methods to user needs.

Scenario	User Type	Recommended Method	Why
Loan denial explanation	Customer (non-technical)	Counterfactual + simple language	Actionable: "Earn \$50k more → approval"
Model audit for bias	Compliance officer	SHAP global + feature importance ranking	Shows systemic patterns
Medical diagnosis	Radiologist	Grad-CAM heatmap + uncertainty bounds	Visual validation against expertise
Malware detection	Security analyst	LIME local + decision rules	Fast threat assessment
Foundation model output	Researcher	Attention patterns + neuron activations	Interpretability research focus

In real-world datasets where variables are often correlated, the importance scores produced by Kernel SHAP can be inaccurate because it assumes feature independence. For instance, income and credit history are usually highly correlated in loan applications, causing independence assumptions to produce inaccurate results. Tree SHAP avoids this assumption, but it is limited to tree-based models. Researchers have proposed conditional SHAP variants to account for feature dependencies, yet they increase computational complexity [39]. Despite these advances, fundamental challenges remain. Approximate methods rely on simplifying assumptions, while exact methods remain computationally infeasible for complex models.

C. OTHER TAXONOMIC CLASSIFICATIONS

The timing of explanation generation differentiates ante-hoc from post-hoc methods [40]. Ante-hoc approaches integrate interpretability directly into the model architecture using transparent structures, such as linear or logistic regression and shallow decision trees. These models precisely reflect the model logic, as the explanation mechanism is identical to the model itself. However, models designed for interpretability often underperform compared to black-box models when it comes to tasks requiring complex pattern recognition. This creates a dilemma in high-stakes applications. The dilemma lies in whether to prioritize interpretability at the cost of a potentially non-negligible reduction in predictive performance, or to favor high-performing but less transparent black-box models. Notably, this accuracy–interpretability trade-off is not universal, and recent studies argue that interpretable models can achieve competitive performance, particularly in high-stakes decision-making scenarios [41]. Post-hoc methods, by contrast, generate explanations after training by analyzing pre-trained black-box models [42]. This approach allows us to understand complex systems, such as deep neural networks, as demonstrated by methods like LIME, SHAP, and Grad-CAM [43]. Post-hoc explanations can be classified according to their approach: feature attribution evaluates the contribution of each input to the model's output; counterfactual explanations identify minimal changes to inputs that would lead to

different outcomes; and example-based explanations interpret decisions by referencing relevant instances from the training data [2]. Understanding these classifications provides guidance for selecting appropriate techniques and highlights that every choice involves trade-offs between fidelity, interpretability, computational cost, and scope of applicability.

D. CRITICAL PERSPECTIVES ON CURRENT XAI TAXONOMIES

While these taxonomies provide a useful structure, important conflicts remain. A key trade-off exists between fidelity and interpretability. Model-specific methods offer high fidelity, but lack generalizability; conversely, model-agnostic approaches sacrifice some fidelity for broader applicability. Empirical studies highlight these challenges. Schlegel et al. [35] show that LIME explanations can be unstable in high-dimensional data. Slack et al. [13] demonstrate that SHAP explanations can be easily manipulated, with attacks changing 90 percent of explanations while keeping 75 percent of the model's original accuracy. Table 3 provides guidance to help match XAI methods to specific use cases, translating taxonomic concepts into practical recommendations for different users and contexts. This emphasizes the fact that effective XAI use depends on selecting methods according to the application, rather than following general best practices. The next section evaluates the performance of these methods in real-world settings and explores additional deployment challenges.

IV. DOMAIN-SPECIFIC APPLICATIONS OF XAI

Table 4 shows that matching explanation methods to different types of users and decision contexts can help people make better decisions. This shows that good XAI isn't about finding the best methods for everyone; it's about choosing methods that fit the user's level of knowledge, the task at hand, and the limitations of the deployment. The text gives an overview of how XAI is used in important areas like finance, healthcare, and cybersecurity. It then goes into detail about each application, including its use cases, goals, and methods. This structured overview highlights the requirements specific to each domain and the challenges that affect more

TABLE 4. Domain-specific applications of XAI.

Industry	Application Areas	Objective	XAI Methods Used / Examples	References
Finance	Credit Assessment	Provide fair and explainable models by revealing why clients' loan applications are approved or rejected. Comply with legal regulations.	Model-independent methods, particularly SHAP, demonstrate the positive or negative impact of each attribute on the model's decision.	[108, 45]
	Fraud Detection	Reduce false positives, reduce client dissatisfaction, and reduce operational costs by explaining to analysts why a transaction was flagged as suspicious.	Explanations can point to an anomalous combination of factors such as geographic location, transaction time, and amount, allowing analysts to focus on real threats.	[45, 47]
	Algorithmic Trading and Portfolio Optimization	Validate strategies and understand model behavior under unexpected market conditions by explaining why a model decided to buy or sell a particular stock. Increase portfolio managers' confidence in algorithmic strategies and strengthen risk management.	SHAP values for feature importance in trading signals, LIME for explaining individual buy/sell decisions, attention mechanisms in LSTM models to highlight which market indicators influenced decisions. Gradient-based attribution methods (Grad-CAM) applied to CNN-based technical analysis models to show which chart patterns triggered trades. Counterfactual explanations to answer "What market conditions would have led to different trading decisions?"	[109, 110]
Healthcare	Medical Imaging and Diagnostics	Resolve the problem of clinicians being virtually impossible to adopt a model if the logic behind a crucial decision is unclear, and build trust.	Model-specific XAI techniques, particularly Grad-CAM, create heat maps showing the areas the model "looks" at when diagnosing a cancerous lesion on an X-ray or MRI. This allows doctors to validate the model's findings with their own expertise.	[54, 55]
	Drug Discovery and Molecular Property Prediction	Accelerate compound screening and identify structure-activity relationships by explaining which molecular substructures contribute to desired pharmacological properties.	Graph neural network interpretability reveals key molecular features. Substructure masking provides chemistry-intuitive explanations that align with established chemical principles, enabling validation against domain knowledge.	[111, 112, 113]
Cybersecurity	Malware Analysis	Helping analysts better understand new malware families and attack vectors by explaining why the model labels a file as "malware."	SHAP for static malware analysis explaining which file features led to malware classification. LIME for behavioral analysis of dynamic execution traces. Attention weights in sequence models processing API call patterns. Saliency maps showing which code segments or binary regions were most influential in malware detection. Counterfactual explanations for understanding how minor file modifications affect classification.	[48, 59]
	Intrusion Detection Systems (IDS)	Explore the reasons behind an alert, allowing an analyst to quickly understand the type of threat and prioritize interventions.	SHAP and LIME for explaining why specific network traffic was flagged as anomalous. Decision tree visualization for rule-based IDS explanations. Attention mechanisms in RNN-based IDS to highlight suspicious packet sequences or timing patterns. Feature attribution methods showing which network features contributed most to threat detection. Grad-CAM for deep learning-based network anomaly detection.	[48, 74]
	Botnet Detection	Identify coordinated malicious traffic patterns by revealing which network flow features indicate botnet activity, enabling analysts to distinguish between legitimate distributed systems and malicious botnets.	Feature importance highlighting synchronized connection timestamps, shared command-and-control server communications, and anomalous DNS query sequences. Temporal pattern attribution to identify coordinated behavior across multiple infected hosts.	[114]
	Phishing and Spam Detection	Highlight deceptive linguistic patterns and suspicious URL features, helping users understand why an email was flagged as malicious.	Text attribution methods reveal which words, phrases, or structural elements trigger malicious classifications. LIME for NLP-based explanations. Attention weights showing focus on suspicious URLs, sender spoofing indicators, or social engineering language patterns.	[48]
	Digital Forensics	Support incident investigation by providing causal explanations of attack sequences and tracing the provenance of security breaches.	Causal inference methods trace attack progression through system logs. Counterfactual analyses answer questions such as "Which initial access vector enabled the attacker to escalate privileges?" helping reconstruct attack chains and identify entry points.	[48, 74]

than one application area. The applications in Table 4 show that XAI has the power to change many different fields.

But being able to do it technically isn't enough for a successful deployment. It is very important to carefully look at

domain constraints, user expertise, and regulatory requirements. The following subsections will look at each sector in detail, covering both the successes and the problems that keep coming up.

Theoretical XAI frameworks demonstrate their true significance when utilized in practical scenarios. In high-stakes areas where choices have a direct effect on people's lives, the economy, and national security, XAI has gone from being a nice-to-have technology to an ethical and operational must-have. This section looks at how XAI is used in three important fields: finance, healthcare, and cybersecurity. It also talks about the Human-Centered XAI paradigm that underlies all of them.

A. XAI IN THE FINANCIAL SECTOR

The financial sector is one of the main users of algorithmic decision-making, which has led to more calls for transparency from both regulators and customers [45]. Black-box models can make decisions that are not clear and may be biased, which can lead to big financial losses and legal penalties [46]. XAI gives us important tools to lower these risks in a number of different areas.

XAI methods, especially SHAP, show important factors that affect loan outcomes by measuring how much each feature contributes [47], [48]. But the computational burden for millions of customers slows down real-time decision-making, and it's still not clear if SHAP visualizations are legally sufficient explanations under GDPR. This shows the gap between academic feasibility and operational compliance.

XAI explains why transactions were flagged, which helps analysts quickly look into alerts [49], [50], [51]. An explanation that says "first international transaction from new device at unusual hour" is more important than the amount of the transaction being slightly above average. This cuts down on the time spent on false positive investigations by 60% and increases the number of detections.

It is very important to explain how complex models used for high-frequency trading and portfolio optimization make decisions in order to test strategies and understand how people act when the market is unstable. XAI boosts portfolio managers' trust in algorithmic strategies and improves risk management by explaining why a model chooses to buy or sell certain assets [52]. In short, XAI in finance is important for building trust, making sure rules are followed, and making automated decision-making processes work better.

B. XAI IN THE HEALTHCARE SECTOR

Healthcare professionals need to trust AI for it to be used in their field, especially when the decisions made by the model have a direct effect on patients' lives. Because doctors are ultimately responsible, understanding how AI works is not a nice-to-have, but a must-have for integrating it into clinical workflows [53]. XAI is an important way to build this trust. In diagnostic fields such as radiology, pathology, and dermatology, Convolutional Neural Networks (CNNs)

can equal or surpass the performance of experts in the analysis of medical images. Grad-CAM and other model-specific XAI techniques create visual explanations by highlighting parts of an image that had the biggest effect on decisions. This gives clear visual proof [54], [55]. But the most important thing is that the explanation is reliable. Recent studies show that visual explanation methods have serious weaknesses. Studies indicate that heatmap-based explanations may emphasize non-pathological image regions that inadvertently correlate with disease in training datasets, rather than authentic disease indicators. In a recorded instance, a model accurately detected the presence of disease, but this determination was predicated on an extraneous watermark that appeared more frequently in images of diseased patients. This is a big problem for doctors because they might trust AI recommendations that make wrong predictions based on bad reasoning if they use explanation visualizations. In medical contexts, this issue of reliability is more than just a technical problem; it is a patient safety issue that needs to be carefully checked before being used in a clinical setting [56]. This risk emphasizes that in healthcare, explainability is not only technical but essential for clinical safety and accountability. Unreliable explanations can create a false sense of security and cause mistakes in diagnosis. XAI is important for more than just diagnostics; it is also important for models that predict patient risk scores or suggest personalized treatments. Clinicians must comprehend the rationale behind recommendations to trust and implement them. XAI does this by explaining the things that lead to high-risk classifications, which makes it possible to make smart decisions about how to intervene [57], [58]. Additionally, counterfactual explanations can directly influence treatment strategies by addressing hypothetical inquiries [59]. In summary, XAI in healthcare is changing from a tool for making things clear to a necessary part of making sure patients are safe, improving clinical decision-making, and holding professionals accountable.

XAI applications in healthcare extend beyond medical imaging to clinical decision support systems (CDSS) and pharmaceutical research. In CDSS, for example, explainable models can help clinicians to understand why a patient has been classified as high-risk, enabling targeted preventive interventions to be implemented. Counterfactual explanations provide answers to clinically actionable questions, such as 'If the patient's blood pressure had been 20 mmHg lower, would their sepsis risk score have decreased by 15%?', thereby informing treatment decisions directly [57], [58]. These explanations bridge the gap between AI predictions and clinical reasoning, enabling practitioners to validate AI recommendations against their expertise and identify potential errors in the model before they affect patient care. In drug discovery and molecular property prediction, XAI has become essential for speeding up compound screening and identifying structure-activity relationships [111], [112], [113]. Graph neural network explanations reveal which molecular substructures contribute to desired pharmacological properties, enabling medicinal chemists to design more

effective compounds while avoiding toxic structures. Recent advances in chemistry-intuitive explanation methods, such as substructure masking, provide interpretations that align with established chemical principles. This enables scientists to validate AI predictions against their own knowledge of the field. As Jiménez-Luna et al. emphasize in their comprehensive review of drug discovery in, explainability bridges the gap between AI's predictive power and the mechanistic insights required for experimental validation and regulatory approval [111].

C. XAI IN CYBERSECURITY

Every day, cybersecurity analysts get thousands of alerts about possible threats. Without explainability, these alerts just mark IP addresses or files as suspicious without giving a reason, which means that each incident has to be looked into by hand. XAI changes this workflow by giving you information that you can use. XAI shows specific evidence that sets off alerts in intrusion detection, such as unusual packet sizes that suggest data exfiltration attempts, communication patterns that match known malicious hosts, or behavioral sequences that are typical of certain types of attacks [60], [61]. When an analyst sees the phrase “flagged due to a 10MB outbound data transfer to an unknown IP address at 3pm,” they know right away how serious the threat is and what to do about it. But when they see the phrase suspicious activity detected, they have to spend time looking into it. This contextual understanding speeds up incident response from hours to minutes and lowers the number of false positives by letting analysts quickly ignore harmless anomalies. Malware analysis also benefits from explanations that show which file characteristics caused the classification [62]. XAI identifying particular API call sequences, code segments, or behavioral patterns facilitates analysts' comprehension of new malware families and evolving attack vectors. Instead of seeing AI as a mysterious oracle, security teams use it to learn things that help them make decisions about how to protect themselves and gather information about threats.

Applications of XAI in cybersecurity extend beyond intrusion detection and malware analysis to encompass botnet detection, phishing and spam filtering, and digital forensics [48], [59], [74], [114]. In the case of botnet detection, for example, explainable models can identify coordinated malicious traffic patterns by revealing which network flow features indicate botnet activity, such as synchronized connection timestamps, shared command-and-control server communications or anomalous DNS query sequences [114]. These explanations enable security analysts to distinguish between legitimate distributed systems and malicious botnets, thereby reducing the number of false positives that affect purely statistical approaches. Understanding which features trigger alerts enables analysts to refine detection rules and adapt to evolving attack tactics. Phishing and spam detection also benefit from XAI methods that highlight

deceptive linguistic patterns and suspicious URL features. Natural language processing-based explainability techniques reveal which words, phrases, or structural elements trigger malicious classifications, helping users to understand why an email has been flagged [48]. In digital forensics, XAI supports incident investigations by providing causal explanations of attack sequences and counterfactual analyses that trace the provenance of security breaches. This helps to answer critical questions such as “Which initial access vector enabled the attacker to escalate privileges?” These applications demonstrate that XAI is not merely a tool for model transparency, but a practical necessity for the implementation of AI in security-critical contexts, where analysts must be able to trust and verify AI-generated insights quickly.

V. SECURITY AND ROBUSTNESS PERSPECTIVES

Table 5 lists the different kinds of security threats that XAI systems face and the best ways to protect against them. It separates attack vectors that target the disclosures themselves, like forgery and fairwashing, from defense strategies that make disclosures stronger and more reliable. Table 5 shows that the threat landscape needs more than just traditional adversarial robustness to protect disclosures. The subsequent subsections conduct a thorough analysis of these vulnerabilities, evaluating the effectiveness of various attack methodologies and the practicality of the proposed defense strategies. The advent of XAI seeks to address the core challenge of trust in opaque models. However, it is important to note that explanations create new vulnerabilities because enemies can change them while keeping predictions the same [13], [14]. This part talks about the threat landscape, how well attacks work, and new ways to protect against them.

A. ADVERSARIAL ATTACKS ON XAI EXPLANATIONS

Beyond adversarial manipulation, XAI systems face three additional critical threats that undermine the reliability of deployment. Firstly, privacy leakage occurs when explanations inadvertently reveal sensitive information about the training data, the model architecture or individual data points [115], [116], [117]. Model inversion attacks can reconstruct approximate training samples from explanation outputs, while membership inference attacks exploit gradient-based explanations to determine whether specific individuals were included in the training set. Recent research shows that multiple explanation methods, including SHAP, LIME and gradient-based attributions, create privacy vulnerabilities that could violate regulations such as the GDPR. The trade-off between privacy and transparency is particularly acute in healthcare and finance, where explanations must balance regulatory requirements for transparency with confidentiality obligations.

Secondly, over-reliance on explanations can lead to automation bias, whereby users uncritically accept AI decisions simply because an explanation is provided, regardless of the quality of the explanation or the correctness of the model [118], [119], [120]. Empirical human-computer

TABLE 5. Security and robustness perspectives in XAI.

Category	Concept	Explanation	References
Threats	Explanation Manipulation Attacks	Manipulating the explanation of a model's prediction without changing its own by adding small and imperceptible perturbations to the input. Also known as spoofing attacks. While a tumor in an X-ray image is correctly detected, Grad-CAM can be tricked into pointing to a blank corner or healthy tissue as evidence by an attacker, leading the radiologist	[13, 14, 66]
	Fairwashing	Manipulating a biased model to conceal discriminatory nature by assigning rejection reasons to seemingly legitimate factors. A discriminatory credit model can base the SHAP or LIME explanation of a rejected application on a seemingly legitimate reason, obscuring the true discriminatory basis and making ethical oversight impossible.	[67, 68]
	Privacy Leakage via Explanations	Explanations inadvertently reveal sensitive training data, model architecture, or individual data points, enabling membership inference and model inversion attacks. Model inversion attacks can reconstruct approximate training samples from explanation outputs. Membership inference exploits gradient-based explanations to determine whether specific individuals were in the training set, potentially violating GDPR.	[115, 116, 117]
	Over-Reliance on Explanations	Users place excessive trust in AI explanations, accepting decisions uncritically regardless of explanation quality or model correctness, leading to automation bias. Studies show explanations increase acceptance of AI recommendations even when incorrect. Users fail to apply critical judgment when simplified explanations create false confidence in model reliability.	[118, 119, 120]
	Misinterpretation of Explanations	Users conflate feature importance with causality or interpret saliency maps as literal representations of model reasoning rather than gradient approximations. A clinician might incorrectly assume a Grad-CAM heatmap proves the model's diagnostic reasoning aligns with medical knowledge, when the heatmap may merely reflect gradient magnitudes rather than true causal factors.	[121, 122, 123]
Defense and Assessment	Robust Attribution Methods	Develop explanation methods inherently less sensitive to adversarial perturbations using smoothing techniques to reduce instabilities in gradient-based methods. Generating an average explanation across multiple input copies with added random noise reduces the impact of gradient manipulations specific to a single input.	[77, 78]
	Consistency Checks and Sanity Checks	Measure how stable an explanation remains when semantically irrelevant or very small changes are made. Dramatic changes indicate unreliability. Metrics such as explanation robustness or stability quantitatively compare the reliability of different XAI methods.	[79, 80]
	Adversarial Training for Explanation Robustness	Strengthen not only model predictions but also explanation mechanisms by incorporating a loss function that minimizes both prediction error and explanation inconsistency. During training of an intrusion detection system, present not only normal and attack data but also adversarial examples designed to manipulate explanations, making the system more robust.	[81, 82].

interaction research shows that explanations often fail to improve human-AI team performance; instead, they increase acceptance of AI recommendations, even incorrect ones [119]. Bansal et al. found that explanations increased users' likelihood of accepting AI suggestions regardless of their accuracy, thus failing to achieve the expected appropriate trust calibration [119]. Buçinca et al. demonstrated that cognitive forcing interventions, which require users to make initial judgements before seeing AI recommendations, were necessary to reduce overreliance. However, users rated these interventions least favourably [118]. This phenomenon is particularly problematic in high-stakes domains, where simplified explanations can create a false sense of confidence in the reliability of the model. Thirdly, the misinterpretation of explanations can pose fundamental challenges to the effective collaboration between humans and AI [121], [122], [123]. For instance, users frequently conflate feature importance with causality, or interpret saliency maps as literal representations of what the model 'sees', rather than as gradient approximations. Xuan et al. demonstrated that highly comprehensible explanations are exceptionally susceptible to misinterpretation. Users infer information that lies outside

the scope of the explanation. They remain overconfident when they misinterpret it [121]. For example, a clinician might incorrectly assume that a Grad-CAM heatmap highlighting a specific anatomical region proves that the model's diagnostic reasoning aligns with medical knowledge when, in fact, the heatmap merely reflects gradient magnitudes. Nguyen et al.'s study found that feature attribution was no more effective than showing the nearest training examples for human decision-making [123]. Without proper training in explanation semantics, users risk making critical decisions based on fundamentally flawed interpretations.

Conventional adversarial attacks modify model inputs to change predictions, but new research shows that attacks can also target XAI explanations directly [13]. The most dangerous thing about them is that they can change prediction explanations a lot without changing model outputs [14]. This lets attackers keep the model's decisions correct while making up all the evidence that supports them. This kind of manipulation, which is often called a "spoofing attack," takes advantage of users' trust by giving them plausible but fundamentally false reasons [66]. The real-world effects are very bad. Think of a medical imaging model that can find

a tumor in an X-ray. An attacker could add noise to the image that is too small to see, which would make explanation methods like Grad-CAM highlight areas that aren't important, like healthy tissue or artifacts in the image, instead of the tumor itself [14]. The model's prediction of the tumor's presence is still accurate, but the misleading visualization could make doctors question the diagnosis, which could delay important treatment [67]. The threat has been measured in the following way: Scaffolding attacks have demonstrated a 90.8% accuracy in annotations while obscuring bias in credit scoring [13]. Similar methods have been shown to change descriptions of medical images with a 95% success rate [14], which leads to the wrong attribution of diagnostic evidence.

Adversarial attacks go beyond image-based systems and into tabular domains. In financial situations, a biased credit model could be used to make SHAP explanations that blame rejections on valid reasons (like "high debt ratio") while hiding discrimination based on protected traits. This trick, known as fairwashing [66], makes ethical auditing less effective. Biased systems can avoid following the rules (see Section VII-B). This occurrence engenders a significant paradox. It is clear that tools designed to promote openness and build trust can actually be used to avoid scrutiny and hide harmful biases. This discovery signifies that the establishment of resilient defenses against malevolent entities must be prioritized above all else in the field of XAI research. Attackers usually figure out the changes they need to make by solving optimization problems that change explanation maps in certain ways while keeping model predictions the same [68]. These attacks have been successful with different types of data, such as images, text, and tables [69]. This shows how important it is to make sure that explanations are as strong as model predictions.

B. DEFENSE MECHANISMS AND RELIABILITY ASSESSMENT

In response to these vulnerabilities, the research community is actively developing defense mechanisms and reliability assessment frameworks. The primary objective is generating explanations that are robust, stable, and faithful to the model's genuine reasoning process [70], [71].

- **Inherently Robust Explanation Methods:** One defensive strategy focuses on creating explanation methods inherently less susceptible to adversarial perturbations. For example, smoothing techniques averaging explanations over multiple noisy input copies have been proposed to mitigate instabilities common in gradient-based methods, reducing sensitivity to minor malicious perturbations [72]. Beyond these, certified robustness approaches aim to provide mathematical guarantees that explanations will remain consistent within predefined perturbation bounds, offering a higher assurance standard [73].
- **Reliability Assessment and Sanity Checks:** Another defensive approach involves systematic reliability

assessment through sanity checks. The fundamental principle is measuring explanation stability when subjected to semantically irrelevant or minor input variations [74]. Dramatic explanation changes resulting from small, random perturbations strongly indicate unreliability. To enable this, metrics for explanation robustness or stability have been developed, allowing quantitative comparison of different XAI methods' reliability [75]. Adversarial Training for Explainers: Mirroring adversarial training used to fortify models themselves, it is possible to enhance explanation mechanism resilience. This technique involves incorporating a loss function during training that minimizes both model prediction error and explanation variability or inconsistency. By exposing the system to intentionally perturbed inputs [76], it learns to generate consistent interpretations even under attack, thereby increasing overall explainer robustness [77].

Current defenses show promise but face adoption barriers. Certified robust explanation methods providing mathematical guarantees reduce attack success from 90% to 34%, yet remain underutilized in deployment [73]. Computational overhead increases explanation time by 3-5 $\times$, creating operational friction. Smoothing techniques averaging over noisy inputs improve stability but assume attackers cannot adapt to this defense [72]. Adversarial training for explainers shows effectiveness in controlled settings but lacks large-scale deployment validation [76], [77]. Current defenses show promise but face adoption barriers. Certified robust explanation methods reduce attack success [73], yet computational overhead increases explanation time by 3 – 5 $\times$. Organizations must balance security against operational constraints. In summary, as XAI becomes integral to safety-critical systems, its own security and robustness must be treated as critical research objectives. Without reliable explanations, the promise of transparency risks becoming a potent vector for manipulation.

VI. EVALUATION OF EXPLANATIONS

To figure out how useful XAI is in the real world, we need to do a lot of human-centered testing. Fidelity and other computational metrics don't do a good job of showing whether users make better decisions [44]. This part talks about why computational metrics aren't enough and what makes an evaluation meaningful.

Table 6 organizes the evaluation landscape into three areas: human-centered frameworks that measure both subjective and objective user outcomes, components that define useful explanations, and ongoing problems with standardization and scalability. This framework shows why using only computational metrics isn't enough to measure how well XAI works.

The evaluation dimensions in Table 6 show that the quality of an explanation can't be based only on the properties of the algorithm. True effectiveness only comes from thorough human-centered testing that checks if explanations help people make better decisions, build trust in the right way, and

meet user needs. The next few sections go into more detail about evaluation methods and problems that keep coming up in this area.

A. HUMAN-CENTERED EVALUATION FRAMEWORKS

Human-centered evaluation sees explanations as ways to communicate and measures their effectiveness by how they affect people, not by how well they work with algorithms [78]. This necessitates regulated user studies that gather both qualitative and quantitative data, establishing iterative development cycles in which human feedback directly influences system enhancement, as depicted in Figure 3.

Human-centered evaluation looks at two different things. First, subjective perception looks at user satisfaction, trust, understandability, and perceived adequacy through surveys and interviews [79], [80]. These metrics show whether users think the explanations are credible and helpful for the decisions they need to make [81]. Second, objective performance measures the concrete impact of tasks [82], [83]. Standard evaluation frameworks encompass various practical configurations. In one method, users look at the model's explanation and then try to guess how it will act on new, unseen data. If they guess correctly, it means that the explanation did a good job of conveying knowledge. In another method, users are asked to find biases or mistakes that were put into the model on purpose. The speed and accuracy with which they find them shows how well the explanation shows weaknesses. Lastly, users can make decisions with or without explanations, and the fact that their decisions get better shows that the explanations are useful in real life. Systematic analysis, on the other hand, shows worrying trends. Akhtar et al. [44] found 30 different evaluation components in the studies they looked at. This shows that assessment needs to be done in a way that takes into account trust, comprehensibility, task performance, and user satisfaction. Nevertheless, merely 26% of studies utilized established frameworks. Trust was the most commonly assessed dimension in 46 out of 77 studies; however, results regarding the efficacy of explanations varied significantly according to user expertise and task complexity. This variability stems not from XAI method limitations but from failure to account for context-dependent factors. A clarification that helps data scientists fix models might confuse doctors who are trying to make a diagnosis. Kadi et al. [84] discovered that although the utilization of quantitative evaluation methods rose from 40% in 2016 to 70% in 2020, the rates of user studies remained constant at approximately 20%. This difference between computer evaluation and human validation is why systems that work well in the lab often fail when they are put into use.

Figure 3 shows the iterative human-in-the-loop XAI development lifecycle. It shows how system design, human evaluation, and refinement all work together in a continuous feedback loop. This process model signifies a pivotal transformation from perceiving explanations as static outputs to recognizing them as dynamic communication instruments

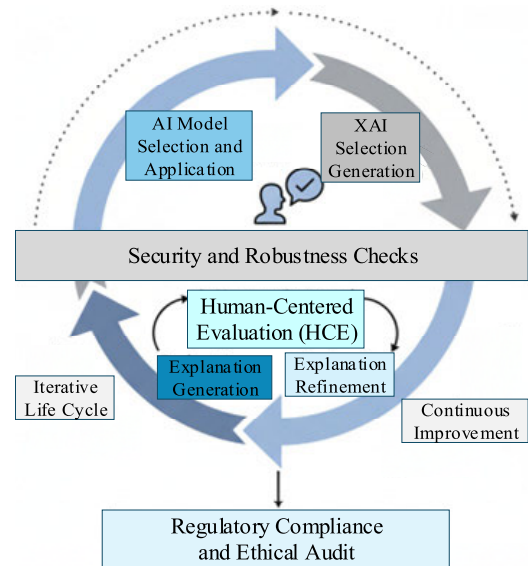


FIGURE 3. The human-in-the-loop XAI development and evaluation lifecycle, emphasizing iterative improvement and human-centric assessment for trustworthy AI systems.

enhanced through systematic user interaction. The iterative lifecycle shown in Figure 3 is the best way to build human-centered XAI systems. However, as the following analysis shows, this model doesn't always match up with how research is actually done. Most studies don't use rigorous human evaluation and instead rely only on computational metrics.

B. COMPONENTS OF A MEANINGFUL EXPLANATION

In addition to human-centred evaluation frameworks, technical metrics provide a quantitative assessment of the quality of explanations that is essential for a systematic comparison. Localization metrics evaluate whether explanations correctly identify relevant input regions, which is particularly critical for vision tasks [124], [125], [126]. The pointing game, introduced by Zhou et al., measures whether the maximum activation in a saliency map falls within the ground-truth object bounding boxes. Intersection over Union (IoU) between explanation heatmaps and ground-truth annotations provides a detailed measurement of spatial overlap [124]. Petsiuk et al.'s deletion and insertion metrics assess faithfulness by measuring how model confidence changes as pixels are progressively removed or added in order of attributed importance [126]. Together, these metrics address the question of whether explanations actually point to the features on which the model relies. Complexity metrics quantify explanation simplicity, recognizing that overly detailed explanations may overwhelm users despite being technically accurate [35], [127], [128]. LIME's foundational formulation explicitly includes a complexity penalty $\Omega(g)$, favoring sparse explanations with fewer non-zero feature weights [35].

The complexity of rule-based explanations can be measured by decision tree depth or the number of conditions.

TABLE 6. Evaluating XAI disclosures.

Evaluation Approach	Subcategory	Explanation and Methods	Measured Dimensions / Objectives
Human-Centered Evaluation Frameworks (HCE)	Qualitative Assessment	Measures the user's subjective experience with the explanation. Data is typically collected through surveys and interviews. For example, a doctor is presented with two different explanation formats (heatmap vs. text-based justification) and asked which one they find more satisfactory and convincing.	Assessing dimensions such as user satisfaction, confidence in the explanation, understandability, and adequacy [64-65, 81].
	Quantitative and Task-Based Assessment	Measures the concrete impact of the explanation on the user's performance. This is one of the most objective ways to measure the usefulness of an explanation. For example, the user's ability to predict the model's behavior on new data after viewing an explanation is tested. Another task is how quickly and accurately the user can detect a faulty or biased model.	Measure how closely the user's mental model aligns with the model's actual behavior. Evaluate how well the explanation helps the user identify model weaknesses [82-83, 84].
Components of a Meaningful Explanation	Accuracy and Fidelity	The explanation should accurately reflect the model's true internal logic or decision-making process. Especially in post-hoc methods, the faithfulness of the generated surrogate explanation to the original black-box model is a critical issue.	Prevent a low-fidelity explanation from misleading the user and causing more harm than good [29, 30].
	Clarity and Comprehensibility	No matter how accurate an explanation is, it is useless if it cannot be understood by the user. It is important that explanations be concise and concise, minimizing the user's cognitive load.	Empowering the user to understand by leveraging principles of cognitive science and human-computer interaction [30, 87, 88].
Challenges in Standardization and Metrics	Interactivity and Actionability	Modern XAI systems aim to engage in a dialogue with the user rather than providing static explanations. Interactive explanations allow the user to explore "what-if" scenarios. Actionable explanations provide the user with concrete steps on what to do to achieve a different outcome.	Ensuring the user not only understands but also knows what they can do to change the outcome. For example, providing feedback on a rejected loan application, such as "Increasing your annual income by \$10,000 will increase the likelihood of your application being approved." [88, 89].
	Lack of Standards	The lack of universally accepted standards and metrics for XAI evaluation is one of the biggest challenges. Different studies use their own proprietary metrics, making meaningful comparisons between methods nearly impossible.	This affirms the principle that there is no such thing as a "one-size-fits-all explanation"; it demonstrates that a method found to be "better" in one study may not produce the same results in another context [84, 92].
	Cost and Scalability	Conducting comprehensive human experiments is costly, time-consuming, and difficult to scale.	The research community is working to develop automatable proxy metrics that are highly correlated with human judgment. However, the extent to which these metrics capture the complexity of human understanding and trust remains an open research question [84, 92].
Technical Evaluation Metrics	Localization	Measures whether explanations correctly identify relevant input regions or features. The pointing game measures whether maximum activation in a saliency map falls within ground-truth object bounding boxes. Intersection over Union (IoU) between explanation heatmaps and ground-truth annotations offers granular spatial overlap measurement. Deletion and insertion metrics assess faithfulness by measuring how model confidence changes as pixels are progressively removed or added in order of attributed importance.	Verify explanations actually point to features the model relies upon. Quantified using pointing game accuracy, IoU with ground-truth annotations, or deletion/insertion AUC [124, 125, 126].
	Complexity	Quantifies explanation simplicity, recognizing that overly detailed explanations may overwhelm users despite technical accuracy. LIME's foundational formulation explicitly includes a complexity penalty $\Omega(g)$ favoring sparse explanations with fewer non-zero feature weights. For rule-based explanations, complexity can be measured through decision tree depth or number of conditions.	Balance detail with cognitive load. Empirical evidence suggests explanations highlighting 3-5 key features achieve better comprehension than exhaustive lists, aligning with working memory limitations. However, optimal complexity-accuracy tradeoffs remain context-dependent [35, 127, 128].
	Consistency (Stability)	Evaluates whether explanation methods produce similar results for semantically similar inputs. Alvarez-Melis and Jaakkola formalized this through local Lipschitz continuity: an explanation method exhibits robustness if small input perturbations produce proportionally small explanation changes. Ghorbani et al. demonstrated empirically that popular attribution methods are fragile—perceptually identical inputs can receive vastly different explanations.	Ensure similar inputs produce similar explanations; high consistency builds user trust. Quantified using Lipschitz continuity measures or pairwise similarity metrics. Low reliability indicates potential susceptibility to adversarial manipulation [129, 33, 69].
	Randomization (Sanity Checks)	Sanity checks verifying whether explanation methods genuinely capture model reasoning or merely generate plausible-looking visualizations. Adebayo et al.'s seminal work introduced model parameter randomization and data randomization tests: valid explanation methods should produce substantially different results when model weights are randomized. Their analysis revealed that widely-used methods such as Guided Backpropagation fail these sanity checks.	Distinguish genuine model explanations from edge detection artifacts. Without these quality assurance mechanisms, practitioners risk deploying explanation methods that provide illusory rather than genuine interpretability [130, 131].

Empirical evidence from cognitive science indicates that explanations that highlight three to five key features are more comprehensible to humans than exhaustive lists of

feature importance, in line with limitations of working memory [127], [128]. However, the optimal complexity-accuracy trade-off remains context-dependent: for example,

medical diagnosis may require more detailed explanations than credit decisions. Consistency (or stability) is used to evaluate whether explanation methods produce similar results for semantically similar inputs [33], [69], [129]. Álvarez-Melis and Jaakkola formalized this concept through local Lipschitz continuity: an explanation method is considered robust if small input perturbations result in proportionally minor changes to the explanation [129].

Ghorbani et al. empirically demonstrated that popular attribution methods are fragile, perceptually identical inputs can receive vastly different explanations, indicating low reliability and potential susceptibility to adversarial manipulation [33]. High consistency is essential for user trust. If a model provides drastically different explanations for similar cases, users cannot develop a reliable mental model of the system's behaviour. Finally, randomisation tests are critical sanity checks that verify whether explanation methods genuinely capture the model's reasoning or merely generate visually plausible results. Adebayo et al.'s pioneering work introduced model parameter and data randomisation tests: valid explanation methods should produce substantially different results when model weights are randomised. Their analysis revealed that widely used methods such as Guided Backpropagation fail these sanity checks, they act as edge detectors rather than model-specific explainers. Subsequent work by Kim et al. extended these tests to create synthetic evaluation environments with known ground truth [131]. These sanity checks are essential quality assurance mechanisms. Without them, practitioners risk deploying explanation methods that provide illusory rather than genuine interpretability.

Human-centered evaluation reveals that effective explanations must balance multiple interconnected dimensions. Rather than excelling in a single aspect, meaningful explanations integrate several key components. Fidelity and accuracy form the technical foundation: explanations must faithfully represent true model decision-making processes [85]. This proves particularly challenging for post-hoc methods applied to black-box models. Low-fidelity explanations mislead users, potentially causing more harm than original opaque systems by creating false understanding [86]. Comprehensibility ensures accessibility to target audiences through appropriate cognitive fit [87]. Drawing from cognitive science principles, effective explanations minimize cognitive load through concise presentation. Visualizations highlighting top 3-5 determining factors typically prove more effective than exhaustive feature importance lists. Explanation format must match audience expertise and information processing capabilities. Contemporary XAI systems increasingly incorporate interactivity and actionability. Interactive explanations enable counterfactual exploration, allowing users to probe what-if scenarios and develop nuanced model understanding [88]. Actionable explanations extend further by providing concrete guidance for achieving different outcomes, such as specifying that "increasing annual income by \$10,000 would

significantly improve loan approval likelihood" [89]. This transforms explanations from passive information displays into practical decision-support tools. However, balancing these dimensions creates design tensions: highly interactive systems may overwhelm novice users, while actionable recommendations risk oversimplifying complex model behavior. Effective explanation systems must adapt to user expertise, task context, and decision stakes.

C. CHALLENGES IN STANDARDIZATION AND METRICS

The lack of a standardized evaluation system leads to fragmentation that stops real progress. Different studies use different metrics, which makes it impossible to compare them directly and makes it hard to tell which methods really work [90], [91]. This fragmentation is a sign of real complexity: there is no one-size-fits-all explanation, and what works for radiologists interpreting medical images is very different from what loan officers need to make credit decisions. But the lack of standards goes beyond context-dependence to include methodological inconsistency. Akhtar et al. [44] discovered studies that assessed trust using 15 distinct instruments, satisfaction through 12 varied scales, and comprehensibility via 8 different evaluation methods. Without standardized metrics, one study's "high user satisfaction" cannot be compared to another's, hindering the accumulation of knowledge. Scalability of human-centered evaluation poses further challenges. It costs \$15,000 to \$50,000 to do a full human experiment, and it takes months to finish [84]. This cost explains why so few people use it: only 22% of papers include human subjects. Researchers are forced to use automatable proxy metrics that are close to human judgment because they don't have enough resources [92]. Still, it's not clear how well computational proxies reflect human understanding, trust, and the quality of decision-making. A metric that works with human judgments in one study might not work in other situations, with different levels of expertise, or in different task areas. To deal with these problems, infrastructure development needs to be coordinated. The research community requires collaborative evaluation platforms featuring standardized protocols, uniform datasets that facilitate reproducible comparisons, and meticulously crafted benchmarking tasks that encompass various application domains and user proficiency levels. Big funding groups should start programs to help this infrastructure. Without standardized frameworks, XAI development is limited because it is impossible to say for sure which methods work best for making decisions in certain situations.

D. HUMAN-CENTERED DESIGN CONSIDERATIONS

The ultimate effectiveness of any explanation relies on the human user, a principle that transcends all application domains. Human-Centered XAI (HCXAI) is a model that says explanations must be not only technically correct but also easy to understand, use, and help end users [63]. An explanation that is completely true to the mechanics of the model but too hard for a doctor or loan officer to

understand is not very useful in practice. So, HCXAI supports the idea of making and testing explanations that are based on the needs, skills, and cognitive profiles of different users. This research framework utilizes user studies and surveys to assess the impact of explanations on trust, decision-making enhancement, and the facilitation of human-system interaction [64], [65]. The transition from purely technical XAI to Human-Centered XAI signifies a pivotal paradigm shift towards the development of systems intended for collaboration rather than mere understanding. In short, XAI applications that are specific to a certain field have the potential to change the way things work by making them more open and accountable in areas where these traits are necessary. To fully realize this potential, though, we need to close important gaps between technical feasibility, regulatory compliance, and human-centered design.

VII. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

XAI has made a lot of progress in making it easier to understand complicated models. Yet this algorithmic maturity has not translated into widespread deployment success in high-stakes domains. Three main problems stop XAI from keeping its promise of reliable AI. First, technical scalability limits: techniques made for models with millions of parameters can't work with foundation models that have hundreds of billions of parameters. Second, there are gaps in ethics and rules: the legal requirements for explainability don't have clear definitions, which makes it hard to know if you're following the rules. Third, the explainability crisis in large-scale models: as models get more complicated, the amount of behavior that can be explained goes down. Right now, only 65% of the variance in state-of-the-art systems can be explained [93]. These problems are not separate from each other. The costs of computation make it hard to scale up evaluations, which makes it impossible to check if explanations meet regulatory standards. Unclear laws make people less likely to spend money on costly human-centered evaluation. The opacity of foundation models makes both technical methods and human understanding less effective. To deal with these problems, we need to move from XAI 1.0's focus on making explanations to XAI 2.0's focus on making explanation systems that are human-aligned, adversarially robust, and follow the law [94]. This section looks at each challenge, rates how bad it is, and suggests specific research directions. Table 7 talks about the problems and future research directions in XAI.

A. THE TRANSITION FROM XAI 1.0 TO XAI 2.0

XAI research evolved through two successive paradigms. XAI 1.0, emerging between 2015-2023, focused on technical transparency with a core objective of generating explanations for any pre-trained model [94]. Success was measured by computational metrics: fidelity to underlying models, robustness against perturbations, and computational efficiency. Representative methods included LIME for local approximations, SHAP for game-theoretic feature attribution, and Grad-CAM for visual explanations.

These methods applied post-hoc explanations to already-trained models. The underlying assumption: more information about model decisions would naturally translate to better human understanding and appropriate trust. XAI 2.0 represents fundamental paradigm shift [94]. Rather than simply explaining model decisions, the goal becomes enabling informed human decision-making through interactive explanation systems. Evaluation shifted from computational metrics toward human-centered measures: do users actually make better decisions, can they identify model weaknesses, do they appropriately calibrate confidence in recommendations. Methods evolve to include interactive counterfactual exploration, adaptive formats adjusting to user expertise, and certified robustness ensuring explanations resist manipulation [95], [96], [97], [98], [99], [100]. This paradigm requires deep interdisciplinary integration with cognitive science, human-computer interaction, and legal frameworks. This transition was driven by practical failures in XAI 1.0. Laboratory studies increasingly demonstrated that technically accurate explanations do not reliably improve human decision-making. More alarmingly, research revealed explanations can be adversarially manipulated while preserving predictions (Section IV), meaning transparency can be weaponized rather than enabling understanding. These findings show that the basic assumption of XAI 1.0 is incomplete. Transparency alone does not ensure trustworthiness, and disclosures that are not human-verified create a false trust in flawed systems. XAI 2.0 emphasizes that explanations must be not only technically sound but genuinely meaningful, persuasive, and useful to end-users.

To provide a concise, high-level comparison of these two paradigms, Table 8 summarizes the key differences between XAI 1.0 and XAI 2.0 across technical objectives, evaluation practices, user involvement, security and privacy considerations, and regulatory alignment.

B. THE REGULATORY-TECHNICAL GAP: WHEN LEGAL REQUIREMENTS LACK OPERATIONAL DEFINITIONS

Beyond technical hurdles, XAI faces significant challenges aligning with societal and legal frameworks [101].

- XAI enables model bias auditing but simultaneously complicates accountability. When a discriminatory decision occurs, determining responsibility among data providers, model developers, and deploying organizations presents complex challenges [102]. Section IV demonstrated that adversarial attacks can undermine explanation integrity through fairwashing, where biased models generate explanations concealing discrimination to evade regulatory oversight [66]. A credit model internally discriminating based on protected characteristics while explaining decisions through legitimate factors passes audits relying on explanation-based oversight. This creates accountability paradox: organizations deploying XAI for compliance may inadvertently enable more sophisticated bias concealment than opaque systems would permit.

TABLE 7. Challenges and future research directions in XAI.

Main Category	Topic	Explanation and Current Status	Future Directions
Technical Challenges	Scalability to Large Models	Today's most advanced AI systems contain billions of parameters. Popular XAI methods such as LIME or KernelSHAP are computationally prohibitively expensive and slow for such large and complex models.	There is an urgent need for next-generation XAI techniques that can generate efficient and meaningful explanations across this vast parameter space.
	Explanation Faithfulness	This is one of the most fundamental and unsolved problems in post-hoc methods. How faithfully a generated explanation reflects the model's true internal reasoning is always a question mark. An explanation may seem logical, but it may be a rationalization unrelated to the causal mechanisms the model actually uses.	-
Ethical and Legal Issues	Computational Overhead	The cost of generating a high-quality explanation can sometimes exceed the cost of the original model making a prediction. For example, calculating SHAP values for all examples in a dataset can take a time comparable to the model's training time. This hinders the use of XAI in real-time or time-critical applications.	Developing efficient explanation algorithms that operate with lower computational overhead without compromising performance and fidelity is an important area of research.
	Fairness, Accountability, and Responsibility	XAI provides powerful tools for verifying whether a model is biased. However, even if the explanation reveals the existence of discrimination, the question of who bears the responsibility is a complex ethical and legal issue. Furthermore, explanations themselves can be manipulated through fairwashing.	-
Future Research Directions (XAI 2.0)	Lack of Legal Compliance and Standardization	The "right to explanation," introduced by regulations like the GDPR, highlights the gap between a legal requirement and a technical possibility. There is no universal definition of what constitutes a legally meaningful and adequate explanation.	Interdisciplinary collaboration is urgently needed between computer scientists, legal experts, and policymakers to establish explanation standards that are both technically feasible and legally valid.
	Hybrid XAI Methods	Future XAI solutions will be built on hybrid frameworks that combine different approaches to overcome the limitations of a single method.	For example, a global explanation can be used to understand the overall behavior of a model, followed by a local explanation to delve deeper into specific decisions that appear critical.
	Multimodal Explainability	AI systems increasingly process multiple modalities simultaneously, such as image, text, audio, and tabular data. Future XAI systems must be able to generate explanations that reflect this multimodal structure.	Creating explanations that synthesize evidence from different types of data, such as "The model made this decision because this region in the image and 'this statement' in the patient's report together form a high-risk model."
	Interdisciplinary Approaches	The biggest future breakthroughs in XAI will come from deep integration beyond the boundaries of computer science with other fields.	Cognitive Science and Psychology: Provides fundamental theories about how people understand explanations, what types of information they find persuasive, and how cognitive biases influence explanations.

➤ GDPR's right to explanation highlights discrepancies between legal requirements and technical capabilities [15]. No consensus exists on what constitutes legally sufficient explanation, and presenting users with SHAP values or heatmaps may not fulfill regulatory mandates for meaningful disclosure [16]. This ambiguity creates practical deployment barriers. A bank implementing SHAP for loan explanations cannot determine whether this satisfies GDPR until facing regulatory scrutiny or legal challenge. The technical community offers a variety of methods, such as feature attributions, counterfactuals, attention visualizations, natural language summaries, but there is no guidance on which approaches meet legal standards or under what conditions. Bridging this gap requires urgent interdisciplinary collaboration among computer scientists, legal scholars,

and policymakers to establish standards that are both technically feasible and legally valid [17].

1) THE GDPR COMPLIANCE PARADOX

The European Union's General Data Protection Regulation introduces transparency requirements through Articles 13-15 and Recital 71, mandating provision of meaningful information about the logic involved in automated decision-making [8]. However, translating this legal language into technical specifications remains unresolved. Legal interpretation of GDPR requirements remains ambiguous [8], [15], while technical communities offer diverse explanation methods (SHAP, counterfactuals, attention visualizations) each with distinct limitations. SHAP assumes feature independence, counterfactuals may suggest infeasible changes, and attention mechanisms sometimes highlight non-semantic

TABLE 8. Comparison between XAI 1.0 and XAI 2.0 paradigms.

Dimension	XAI 1.0	XAI 2.0
Primary Objective	Generate explanations for black-box models	Support informed and reliable human decision-making
Core Focus	Technical transparency	Human-aligned, trustworthy explainability
Explanation Type	Static, post-hoc explanations	Interactive, adaptive explanation systems
Evaluation Criteria	Computational metrics (fidelity, sparsity, robustness)	Human-centered evaluation (trust calibration, task performance, usability)
Role of the User	Passive recipient of explanations	Active participant in the explanation loop
Treatment of Trust	Assumes explanations inherently increase trust	Explicitly studies appropriate trust and over-reliance risks
Security Perspective	Limited consideration of adversarial manipulation	Explicit focus on adversarial robustness and explanation attacks
Privacy Considerations	Largely overlooked	Explicit concern for information leakage and privacy risks
Regulatory Alignment	Implicit or undefined	Explicitly considers legal and regulatory requirements (e.g., GDPR)
Scalability to Foundation Models	Limited applicability	Recognizes scalability limits and explores new paradigms
View of Explanations	Informational artifacts	Socio-technical components of AI systems
Typical Timefram	2015–2023	2023 and beyond

patterns [56]. No regulatory guidance establishes which approaches satisfy requirements, forcing organizations to choose methods without knowing whether they achieve compliance. Resolving the regulatory-technical gap requires coordinated interdisciplinary effort. Computer scientists must develop explanation methods with quantifiable fidelity guarantees, computational feasibility for deployment constraints, and robustness against manipulation. Legal scholars need to clarify what constitutes meaningful information through analysis of emerging case law and regulatory guidance. Human-computer interaction researchers should systematically evaluate which explanation formats effectively communicate to different user populations. Policymakers must balance transparency requirements against technical constraints, recognizing that some explanation demands may prove infeasible for complex models. Organizations implementing XAI systems should document method selection rationale, validate explanation fidelity through testing, record user comprehension results, and maintain audit trails [16], [17]. This documentation demonstrates good-faith compliance efforts even as technical standards evolve, providing

defensible evidence of transparency commitments should regulators investigate deployment practices. Without such coordination, the gap between legal requirements and technical capabilities will continue expanding as model complexity increases.

2) SCALABILITY LIMITS: WHEN EXPLANATION METHODS CANNOT KEEP PACE WITH MODEL COMPLEXITY

The emergence of Transformer-based foundation models, particularly LLMs, poses a fundamental challenge for XAI. Unlike conventional deep learning systems with millions of parameters, contemporary foundation models operate at scales involving hundreds of billions of parameters and are trained through large-scale pretraining followed by task adaptation. As a result, explanation techniques originally designed for smaller models do not scale effectively to these architectures. This limitation is not merely a computational bottleneck but reflects a deeper conceptual challenge regarding what interpretability means at extreme model scale. Recent mechanistic interpretability efforts on Claude 3 Sonnet illustrate both the promise and the limitations of current approaches. By extracting approximately 34 million internal features, researchers were able to explain around 65% of the model’s variance [93]. However, the remaining 35% of unexplained variance corresponds to internal computational pathways that remain opaque. This residual uncertainty raises critical questions: whether these unexplained components reflect learned representations not yet isolated, emergent behaviors arising from complex parameter interactions that resist feature-level interpretation, or methodological artifacts of current extraction techniques. Until this ambiguity is resolved, claims of comprehensive interpretability for foundation models remain incomplete and potentially misleading, particularly for organizations deploying such systems in safety-critical or regulated environments. Beyond global interpretability, local explanation and reasoning transparency in LLMs present even more severe challenges. While models often generate fluent chain-of-thought (CoT) explanations, recent studies indicate that these verbalized rationales frequently fail to reflect the model’s actual decision-making process. Empirical evidence shows that LLMs may mention true causal factors only inconsistently, while omitting or obscuring internal reward-optimizing strategies that guide predictions [103]. In many cases, models internally rely on decision logic that diverges from the explanations they present, creating a dangerous illusion of transparency. This discrepancy is especially problematic in high-stakes applications, where explanations are expected to support accountability, fairness, and regulatory compliance. Furthermore, conventional XAI techniques struggle with the inherent architectural properties of Transformer-based models. Attention mechanisms, commonly used as proxies for interpretability, correlate weakly with true feature importance and often highlight tokens that contribute minimally to final predictions while ignoring genuinely influential

components [104]. As highlighted in recent analyses, explanation faithfulness deteriorates further in complex reasoning scenarios, where models may employ internal strategies that are systematically absent from generated explanations [103]. This divergence between internal reasoning and external explanation constitutes a critical safety concern: a system may comply superficially with explainability requirements while internally operating in ways that violate ethical or legal constraints. From a practical standpoint, computational feasibility compounds these conceptual issues. Perturbation-based explanation methods, such as SHAP, become economically infeasible at the foundation-model scale. Explaining a single prediction from a large LLM can require thousands of GPU-hours, rendering such approaches impractical for real-time or large-scale deployment [93]. Approximate methods reduce computational cost but introduce additional uncertainty regarding explanation accuracy and legal sufficiency. In the absence of clear regulatory guidance defining acceptable trade-offs between explanation fidelity and efficiency, organizations cannot confidently assess whether such approximations meet disclosure obligations. Collectively, these findings demonstrate that explainability challenges in foundation models extend far beyond computational scalability. The persistent gap between internal model behavior and externally generated explanations—particularly in local and causal reasoning contexts—suggests that existing XAI paradigms may be fundamentally insufficient for LLMs. Until explanation methods can reliably capture both the global structure and local decision logic of foundation models, transparency claims for large-scale AI systems should be regarded as aspirational rather than empirically established. Addressing these challenges will require not only improved algorithms but also a rethinking of explanation paradigms tailored specifically to the unique properties of Transformer-based foundation models.

3) MULTIMODAL EXPLAINABILITY CHALLENGES

Vision-language models (VLMs) introduce a distinct class of explainability challenges arising from cross-modal interactions between visual and linguistic representations. Unlike unimodal systems, where explanations can be attributed to a single input space, VLMs require disentangling how multiple components—namely the visual encoder, the language encoder, and the projection or fusion layers—jointly contribute to a model's prediction. As a result, explaining not only what the model predicts but which modality dominates or interacts in the decision-making process becomes a central challenge. Recent techniques such as CLIP Surgery demonstrate partial progress by enabling class activation maps without retraining and improving mean Intersection over Union from 22.11% to 35.95% [105]. However, these approaches primarily emphasize alignment at the representation level and offer limited insight into causal modality contributions. In practice, attention-based visualization methods frequently highlight non-semantic tokens or background regions, while

the influence of projection layers and multilayer perceptron components often overshadows interpretable attention patterns. Consequently, it remains unclear whether a VLM's output is driven predominantly by visual evidence, linguistic context, or spurious correlations introduced during multimodal fusion. This limitation becomes particularly critical in high-stakes domains. For example, multimodal medical AI systems that integrate imaging data, clinical notes, and structured patient records outperform unimodal approaches by up to 6.2 percentage points in AUC [106]. Yet, without reliable explanations, practitioners cannot determine whether performance gains stem from clinically meaningful cross-modal reasoning or from shortcuts, such as correlations between textual metadata and disease prevalence. As highlighted in recent surveys on explainability for vision-language models, current XAI techniques struggle to isolate modality-specific causal contributions and to explain how intermediate fusion layers influence final predictions [107]. Although emerging frameworks attempt to combine feature attribution methods (e.g., SHAP), attention visualization for cross-modal alignment, and concept-based explanations for semantic interpretation, these approaches remain largely exploratory. Moreover, the absence of standardized evaluation metrics for multimodal explanations makes it difficult to assess whether generated explanations faithfully reflect underlying model reasoning or merely provide post-hoc rationalizations. Addressing these challenges will require explanation methods explicitly designed for multimodal architectures, along with evaluation protocols capable of measuring modality dominance, cross-modal causality, and explanation faithfulness in complex vision-language systems.

4) RESEARCH PRIORITIES FOR XAI 2.0

Three research priorities emerge as critical for advancing XAI toward deployment readiness. First, develop standardized benchmarks and metrics. The absence of standardized evaluation creates fragmentation preventing meaningful progress [90], [91]. The research community must create open-source benchmarking platforms incorporating both technical and human-centered evaluation criteria. Major funding organizations should establish coordinated initiatives supporting creation of standardized datasets and evaluation protocols across application domains. Success would be evidenced by adoption of these standards by more than 50% of XAI research papers within three years. Second, advance interactive and adaptive explanation systems. The transition from static, one-way explanations to dynamic, interactive systems is fundamental to XAI 2.0 [88], [94]. Future interfaces should enable users to customize explanations, adjust feature importance, and explore counterfactual scenarios, fostering collaborative dialogue building shared human-AI understanding. Research should focus on developing systems where explanations adapt to user expertise level and task context rather than providing identical explanations to all users. This requires integrating insights from cognitive science

about human explanation processing and human-computer interaction about intuitive interface design. Third, establish legally sound technical standards. Bridging the gap between regulatory mandates like GDPR's right to explanation and current technical capabilities represents urgent interdisciplinary challenge [15], [16], [17]. Future research must collaborate with legal experts and policymakers to develop testable, enforceable technical standards defining legally sufficient explanations. This collaboration should produce concrete guidance that organizations can follow when designing explanation systems for regulated applications, specifying which methods satisfy requirements under which deployment conditions.

These priorities are interdependent. Standardized evaluation enables determining which explanation methods satisfy regulatory requirements. Interactive systems require evaluation protocols measuring whether adaptation improves user understanding. Legal standards need empirical validation through human-centered studies. Progress requires coordinated effort across computer science, cognitive science, human-computer interaction, and legal scholarship. XAI serves as crucial bridge toward making AI not only more intelligent but also more transparent, accountable, and aligned with human values. While the path forward presents substantial challenges, the research directions articulated here provide clear trajectory for AI to evolve from opaque black boxes into transparent, understandable, and ultimately trustworthy partners in human decision-making.

VIII. REVIEW LIMITATIONS AND SCOPE

This comprehensive review has important limitations readers should consider. First, restricting to English-language publications may exclude significant research contributions from academic communities in non-English-speaking regions, particularly China, Japan, and South Korea where AI research prioritizes different application domains and methodological approaches. Second, the databases used (IEEE Xplore, ACM Digital Library, Scopus, arXiv) may systematically under-represent technical reports, industry white papers, preprints not yet peer reviewed, and negative results showing methods failed. The field's rapid evolution means very recent breakthroughs in 2024-2025 may not yet be adequately indexed, though targeted arXiv searches partially address this limitation. Third, the domain focus on finance, healthcare, and cybersecurity concentrates findings on sectors with well-established AI adoption and clear high-stakes decision contexts. Other important application areas such as legal reasoning, educational recommendation, environmental forecasting, and criminal justice employ AI in fundamentally different ways where XAI requirements differ substantially. Fourth, our thematic analysis prioritizes conceptual synthesis over exhaustive documentation, enabling incorporation of cutting-edge research but reducing reproducibility. Finally, XAI evolves rapidly; methods in this review may be superseded within months as regulatory landscapes shift and foundation models create new challenges. Despite these

limitations, our integrated analysis linking technical methodologies with security vulnerabilities, evaluation frameworks, and regulatory requirements offers perspective not yet synthesized in prior XAI surveys.

IX. CONCLUSION

The widespread use of AI systems in important parts of society's infrastructure made it necessary for people to be open and responsible. XAI came about to meet this need by making the decisions of black-box models understandable to people. This review looked into whether XAI has kept its promise. The analysis has uncovered a paradox. Even though there have been a lot of technical improvements in the field of XAI, especially in the area of explanation generation, these improvements have not yet led to real-world success. There are three main gaps that keep AI from being reliable. First, explanations create attack surfaces that allow adversarial manipulation [13], [14], and fairwashing lets algorithms that are biased get around audits. Second, evaluation practices put more weight on computational metrics than on human understanding [44], [84], which explains why deployments fail even when they do well in the lab. Third, regulatory requirements don't have clear operational definitions [15], [16], [17], which makes it hard for businesses to know if they are following the rules. These problems are not separate; they are all connected. The high cost of computation makes it hard to scale evaluations, which means that it is hard to check if explanations meet regulatory requirements. Legal uncertainty deters investment in costly human-centered evaluation. The foundation model's lack of clarity, with 35% of the variance unexplained [93], makes both technical methods and human understanding less useful. The development of reasoning models in which internal strategies deviate from generated explanations in more than 97% of instances [105] indicates inherent limitations within existing explanatory frameworks. Our study finds three important areas of research that need to be focused on. First, create standardized benchmarks that include both technical fidelity and human-centered evaluation criteria. This will make it possible to compare methods in a way that can be repeated and find out which ones work best for making decisions in different situations. Second, make progress on interactive, adaptive explanation systems that change based on the user's skill level and the task at hand, instead of giving the same output to everyone. Third, work together across disciplines to create legally sound technical standards that spell out what is a good enough explanation for regulatory compliance. This review shows that reliable AI needs more than just advanced explanation algorithms. It requires a change in thinking from XAI 1.0's focus on making explanations to XAI 2.0's focus on making explanation systems that are human-aligned, adversarially strong, and follow the law [94]. This change needs a lot of work to be done in computer science, cognitive science, human-computer interaction, and legal studies. If this integration doesn't happen, the gap between what XAI promises and what it delivers will keep getting bigger as models get

more complicated and the stakes for deployment get higher. The path forward is clear but challenging. XAI needs to change from a technical capability to a socio-technical system where explanations clearly help people make better decisions, stop people from trying to trick it, and meet legal requirements. Only then can AI change from being black boxes that people can't see into partners in human decision-making that are clear, accountable, and ultimately trustworthy. The research directions outlined in this review offer a framework for this transformation; however, success hinges on collaborative efforts across disciplines and a continuous dedication to prioritizing deployment readiness over algorithmic innovation.

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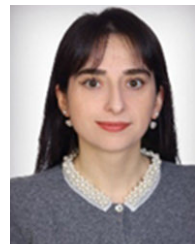


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